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Gaussian Process for Surrogate Modeling  
Covariance Functions

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# Outline

- 1 Eigenfunction Analysis of Kernels
- 2 Numerical Approximation of Eigenfunction
- 3 Piecewise Polynomial Covariance Functions
- 4 Non Stationary Kernel
- 5 Making of New Kernel

# Mercer's Theorem

**Setup:**  $X$  is the input set (domain), not necessarily finite.  $\mu$  is a measure on  $X$  - either Lebesgue measure over a compact  $C \subset \mathbb{R}^D$ , or  $d\mu(x) = p(x) dx$  for a density  $p(x)$ .  $L^2(X, \mu)$  is the Hilbert space of square-integrable functions on  $X$ .

A function  $\phi(\cdot)$  satisfying  $\int k(x, x') \phi(x) d\mu(x) = \lambda \phi(x')$  is an **eigenfunction** with **eigenvalue**  $\lambda$ . There are generally infinitely many, ordered  $\lambda_1 \geq \lambda_2 \geq \dots$ , orthonormal:  $\int \phi_i \phi_j d\mu = \delta_{ij}$ .

**Theorem 4.2 (Mercer).** For  $K_{ij} = k(x_i, x_j)$  positive semi-definite (all eigenvalues non-negative), the eigenvalues  $\{\lambda_i\}$  are absolutely summable and

$$k(x, x') = \sum_{i=1}^{\infty} \lambda_i \phi_i(x) \phi_i^*(x')$$

converges absolutely and uniformly,  $\mu^2$ -almost everywhere - the infinite-dimensional analogue of diagonalizing a Hermitian matrix.

# Degenerate Kernels

**Definition** A **degenerate** kernel has only finitely many non-zero eigenvalues - equivalently, **finite rank**. A kernel that is not degenerate is **nondegenerate**.

**Example.** An  $N$ -dimensional linear regression model in feature space gives rise to a degenerate kernel with at most  $N$  non-zero eigenvalues.

**Finite measure.** If  $\mu$  puts weight on only  $n$  points in  $x$ -space, the eigendecomposition reduces to that of an  $n \times n$  matrix - even for a nondegenerate kernel like the SE kernel.

# Analogy: Covariance Matrix $\leftrightarrow$ Kernel

**Covariance matrix (finite dim.):**

$$\Sigma = \sum_i \lambda_i v_i v_i^\top$$

$\lambda_i$  tells how much variance the data has along direction  $v_i$ .

The same thing happens in function space:  $\lambda_i$  tells how much variance the random function  $f$  has along direction  $\phi_i$  - i.e. how large a weight/coefficient that basis function is allowed to have.

- Small  $\lambda_i \rightarrow$  eigenfunction barely contributes
- Large  $\lambda_i \rightarrow$  huge contribution

**Finite total variance:**  $\sum_i \lambda_i < \infty$ .

**Orthonormal:**  $\phi_i \perp \phi_j$  means non-redundant, independent patterns of variation,  $\langle \phi_i, \phi_j \rangle = \int \phi_i(u) \phi_j(u) du$  (complex conjugate over  $\mathbb{C}$ ).

# Constructing a Random Function from Eigenfunctions

Build a random function as a linear combination of (independent) eigenfunctions:

$$f(u) = \sum_{i=1}^{\infty} w_i \phi_i(u), \quad w_i \sim \mathcal{N}(0, \lambda_i)$$

**Covariance of  $f(u), f(u')$ :**

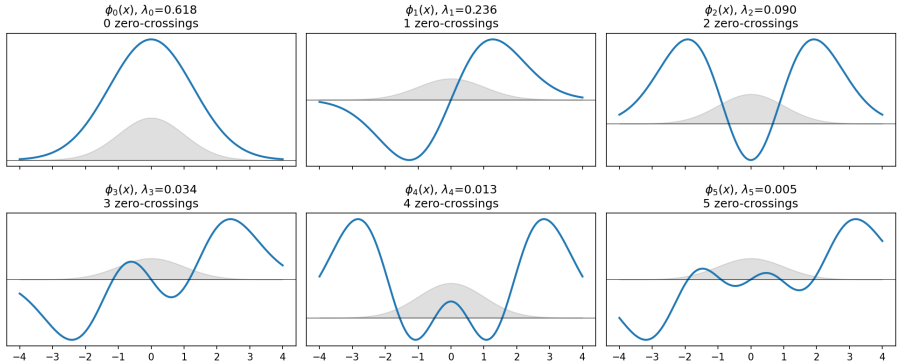
$$\text{Cov}(f(u), f(u')) = \mathbb{E} \left[ \sum_{i,j} w_i w_j \phi_i(u) \phi_j(u') \right] = \sum_{i,j} \phi_i(u) \phi_j(u') \mathbb{E}[w_i w_j]$$

Since  $w_i, w_j$  are independent, zero mean:  $\mathbb{E}[w_i w_j] = 0$  ( $i \neq j$ ),  $\mathbb{E}[w_i^2] = \lambda_i$ .

**Result:**

$$\text{Cov}(f(u), f(u')) = \sum_i \lambda_i \phi_i(u) \phi_i(u')$$

Mercer eigenfunctions of the SE kernel (not the kernel matrix -- the actual basis functions)  
 index  $k$  = number of zero-crossings = how much oscillation that basis direction carries



# Bochner's Theorem & Stationary Kernels

Mercer's theorem assumed a *finite* measure  $\mu$ . Replace it with Lebesgue measure and consider a stationary covariance function; Bochner's theorem gives

$$k(x - x') = \int_{\mathbb{R}^D} e^{2\pi i \mathbf{s} \cdot (x - x')} d\mu(\mathbf{s}) = \int_{\mathbb{R}^D} e^{2\pi i \mathbf{s} \cdot x} \left( e^{2\pi i \mathbf{s} \cdot x'} \right)^* d\mu(\mathbf{s})$$

The complex exponentials  $e^{2\pi i \mathbf{s} \cdot x}$  are the eigenfunctions of a stationary kernel w.r.t. Lebesgue measure. Compare to Mercer's sum: the discrete sum over  $i$  becomes a continuous integral over frequency  $\mathbf{s}$ .

*Limitation of Mercer's theorem:* requires a finite measure space; over unbounded Lebesgue measure, use Bochner's theorem instead.

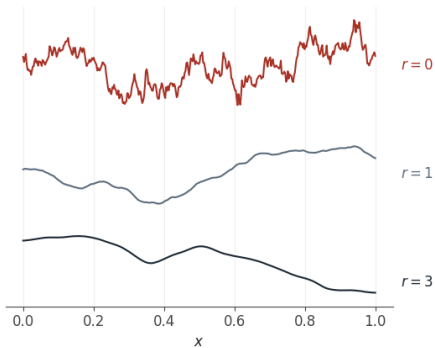
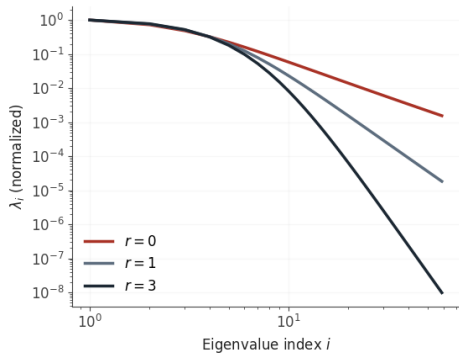
# Eigenvalue Decay and Smoothness

The rate of decay of  $\lambda_i$  encodes the **smoothness** of the kernel/process.

For the specific case of  $\mu$  uniform on  $[0, 1]$ , processes that are  $r$ -times mean-square differentiable have

$$\lambda_i \propto i^{-(2r+2)} \quad \text{asymptotically.}$$

Rougher processes carry more high-frequency power  $\Rightarrow$  their eigenvalue spectrum decays **slowly**. Smoother processes barely need high-frequency terms  $\Rightarrow$  decay is **fast**. Same phenomenon visible in the Matérn power spectrum, eq. (4.15).



# Numerical Approximation: the Nyström Method

Exact solutions of the eigenproblem are rare; the standard numerical routine approximates the integral. With  $x_1, \dots, x_n \sim p(x)$ :

$$\lambda_i \phi_i(x') \approx \frac{1}{n} \sum_{l=1}^n k(x_l, x') \phi_i(x_l)$$

Setting  $x' = x_j$  gives the **matrix eigenproblem**:

$$K u_i = \lambda_i^{\text{mat}} u_i, \quad K_{ij} = k(x_i, x_j), \quad u_i^\top u_i = 1$$

Normalization mismatch  $\Rightarrow \phi_i(x_j) \approx \sqrt{n} (u_i)_j$ , so  $\frac{1}{n} \lambda_i^{\text{mat}}$  estimates  $\lambda_i$ . For fixed  $i$ ,  
 $\frac{1}{n} \lambda_i^{\text{mat}} \rightarrow \lambda_i$  as  $n \rightarrow \infty$  [Baker, 1977, Thm. 3.4]

**Finite-sample bias** (via kernel PCA in feature space), for any  $1 \leq l \leq n$ :

$$\mathbb{E}_n \left[ \frac{1}{n} \sum_{i=1}^l \lambda_i^{\text{mat}} \right] \geq \sum_{i=1}^l \lambda_i \quad (\text{top-}l \text{ sum: } \textit{overestimated})$$

$$\mathbb{E}_n \left[ \frac{1}{n} \sum_{i=l+1}^n \lambda_i^{\text{mat}} \right] \leq \sum_{i=l+1}^{\infty} \lambda_i \quad (\text{tail sum: } \textit{underestimated})$$

[Shawe-Taylor & Williams, 2003]

# Nyström Extension $\equiv$ Kernel PCA Projection

Dividing eq. (4.42) by  $\lambda_i$  extends the approximation from training points to **any** new point  $x'$ :

$$\phi_i(x') \approx \frac{\sqrt{n}}{\lambda_{\text{mat}}} \mathbf{k}(x')^\top u_i, \quad \mathbf{k}(x') = (k(x_1, x'), \dots, k(x_n, x'))^\top \quad (4.44)$$

Same matrix, same eigendecomposition - kernel PCA and Nyström differ only in normalization convention

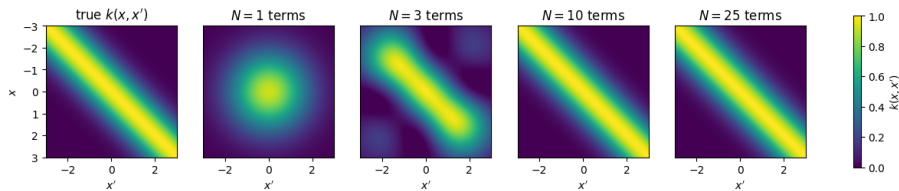
# What Does Mercer's Theorem State?

*(Rank of a matrix = number of non-zero eigenvalues.)*

- 1 Kernels secretly encode an orthogonal basis  $(\phi_i)$  and weights  $(\lambda_i)$ , exactly like a covariance matrix in finite dimensions.
- 2 Eigenvalues measure how much a kernel varies along each basis direction; their decay rate measures smoothness.
- 3 Low rank  $\Rightarrow$  the kernel/process can be approximated smoothly (degenerate).
- 4 **Limitations:** requires a finite measure space; unbounded Lebesgue measure needs Bochner's theorem instead.

**Assumptions:**  $k$  is bounded,  $\mu$  is a finite measure, and  $K$  is PSD  $\Rightarrow$  the eigen-expansion (Mercer's theorem) exists.

Mercer's theorem rebuilding the SE kernel, term by term



# Piecewise Polynomial Covariance Functions

A family of covariance functions with **compact support**: the covariance between two points is exactly zero once their distance exceeds a threshold. This makes the Gram matrix **sparse** by construction - a real computational advantage.

The challenge is guaranteeing **positive definiteness**. These kernels are typically valid only up to some maximum input dimension  $D$ :

$$k_{\text{pp},D,q}(r) = (1 - r)_+^j, \quad j = \lfloor \frac{D}{2} \rfloor + q + 1$$

These kernels are  $2q$ -times continuously differentiable  $\Rightarrow$  the corresponding process is  **$q$ -times mean-square differentiable**. Larger  $q$  trades sparsity for smoothness.

Sparsity is especially useful with iterative GPR solvers (conjugate gradients), where covariance-matrix vector products are the core computation.

# Further Properties: Anisotropy & Periodicity

**Non-monotonic kernels.** Covariance functions needn't decay monotonically or stay positive:  $k(r) = c(\alpha r)^{-\nu} J_\nu(\alpha r)$  is valid for  $\nu \geq (D-2)/2$ ,  $\alpha > 0$  - a *damped oscillation*.

**Anisotropy.** Replace isotropic distance with

$$r^2(x, x') = (x - x')^\top M (x - x'), \quad M \succeq 0.$$

Diagonal  $M \Rightarrow$  different length-scales per dimension (**ARD**). More generally, the **factor analysis distance**  $M = \Lambda \Lambda^\top + \Psi$  (4.22), with  $\Lambda$  a  $D \times k$  matrix of  $k$  high-relevance directions and  $\Psi$  diagonal.

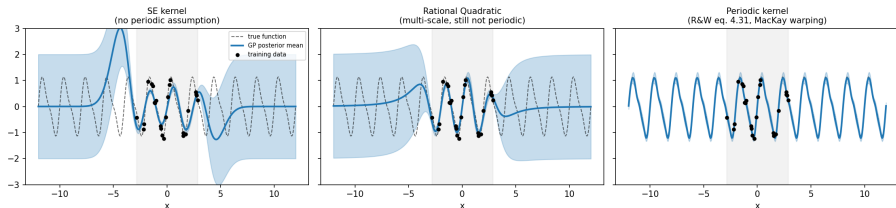
**Periodic kernels.** From any stationary  $k(x)$  on  $\mathbb{R}$ , wrapping it gives a periodic kernel with period  $\ell$ :

$$k_T(x) = \sum_{m \in \mathbb{Z}} k(x + m\ell).$$

**Warping.** Map  $x \mapsto u(x)$ , then use a stationary kernel in  $u$ -space  
 -  $x, u$  need not share a dimension. MacKay's periodic example:  
 $u(x) = (\cos x, \sin x)$  with an SE kernel in  $u$ -space gives

$$k(x, x') = \exp\left(-\frac{2 \sin^2\left(\frac{x-x'}{2}\right)}{\ell^2}\right).$$

Consequence of kernel choice: extrapolating a GP surrogate beyond its training window  
 (gray band = where data actually exists)



# Polynomial (Dot-Product) Kernels

$k(x, x') = \sigma_0^2 + x \cdot x'$  comes from linear regression **homogeneous** if  $\sigma_0^2 = 0$ . Generalizes to  $\sigma_0^2 + x^\top \Sigma_p x'$ , and  $(\sigma_0^2 + x^\top \Sigma_p x')^p$  is valid for positive integer  $p$  (a power of a valid kernel is valid).

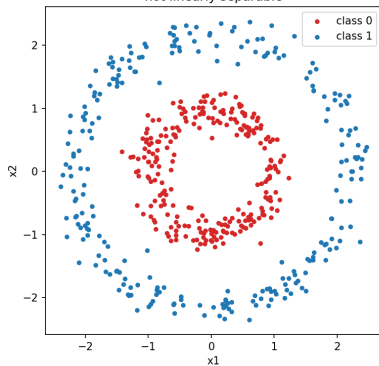
**Explicit feature map.** For  $(x \cdot x')^p = \phi(x) \cdot \phi(x')$ , the feature indexed by monomial exponents  $m$  whose entry  $m_d$  specifies the number of times index  $d$  appears in the monomial (with  $\sum_d m_d = p$ ) is

$$\phi_m(x) = \sqrt{\frac{p!}{m_1! \dots m_D!}} x_1^{m_1} \dots x_D^{m_D}. \quad (4.24)$$

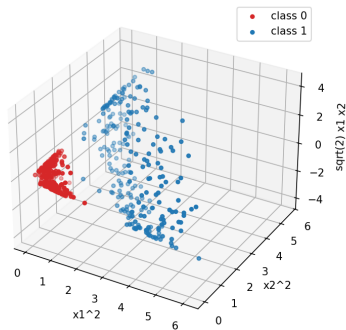
E.g.  $p = 2, D = 2$ :  $\phi(x) = (x_1^2, x_2^2, \sqrt{2} x_1 x_2)^\top$ .

A strange choice for **regression** prior variance grows rapidly with  $|x| > 1$  - but effective for high-dimensional **classification** (e.g. binary/greyscale images normalized to  $[-1, 1]$ ).

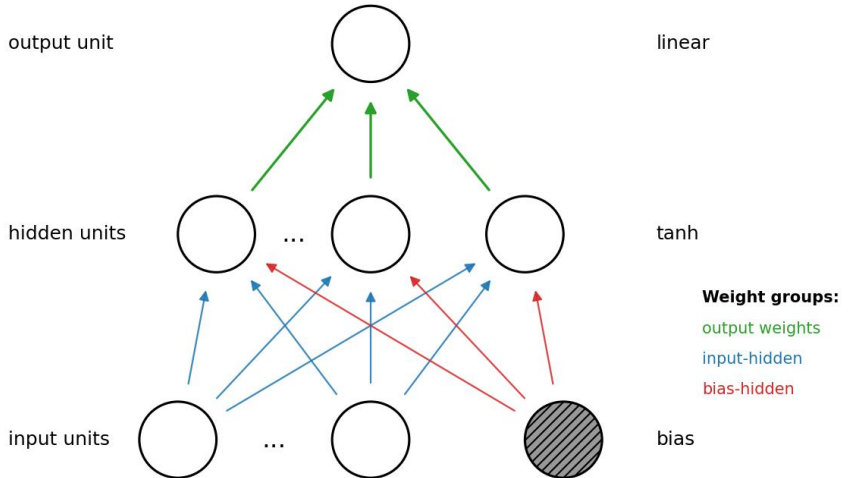
Input space (2D)  
not linearly separable



Feature space:  $\phi(x) = (x_1^2, x_2^2, \sqrt{2}x_1x_2)$   
(R&W eq. 4.24) — now linearly separable



# Feed Forward Neural Networks



A feed forward neural network implements the function: 
$$f(x) = \sum_{i=1}^H v_i \tanh\left(\sum_j u_{ij} x_j + b_i\right)$$

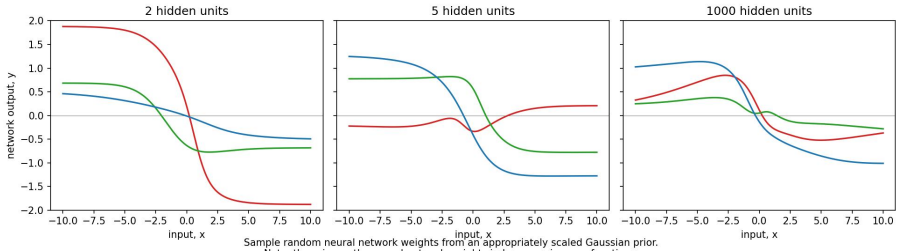
# Neural Network & Warped Kernels

**Neural network kernel:** one hidden layer,  $N_H \rightarrow \infty$  units  $\Rightarrow$  converges to a GP (central limit theorem). With  $h(x; u) = \text{erf}(u_0 + u^\top x)$ ,  $u \sim \mathcal{N}(0, \Sigma)$ :

$$k_{\text{NN}}(x, x') = \frac{2}{\pi} \sin^{-1} \left( \frac{2\tilde{x}^\top \Sigma \tilde{x}'}{\sqrt{(1 + 2\tilde{x}^\top \Sigma \tilde{x})(1 + 2\tilde{x}'^\top \Sigma \tilde{x}')}} \right) \quad (4.29)$$

(the “sigmoid” kernel  $\tanh(a + b x \cdot x')$  is *never* PSD - not a valid kernel.)

## Limits of Large Neural Networks



# Non-Stationary Length-Scales & Other Constructions

**Gibbs' varying-length-scale kernel:** let  $\ell_d(x) > 0$  vary with  $x$  per dimension:

$$k(x, x') = \prod_{d=1}^D \sqrt{\frac{2\ell_d(x)\ell_d(x')}{\ell_d(x)^2 + \ell_d(x')^2}} \exp\left(-\sum_{d=1}^D \frac{(x_d - x'_d)^2}{\ell_d(x)^2 + \ell_d(x')^2}\right) \quad (4.32)$$

Shorter length-scale  $\Rightarrow$  faster-varying samples, as expected;  $k(x, x) = 1$  always.

**Paciorek & Schervish** generalize this to *any* stationary isotropic  $k_S$ , using a matrix-valued  $\Sigma(x) \succ 0$ :

**O'Hagan :** let regression weights  $w$  vary with  $x$  via a GP prior,  $\text{cov}(w(x), w(x')) = W_0 k_w(x, x')$ , giving  $k_f(x, x') = \phi(x)^\top W_0 \phi(x') k_w(x, x')$ .

**Wiener process:**  $k(x, x') = \min(x, x')$  - the fundamental non-stationary kernel.

# Making New Kernels from Old: Sum, Product, Scaling

**Sum.**  $k = k_1 + k_2$  is a valid kernel. *Proof:* take  $f(x) = f_1(x) + f_2(x)$  with  $f_1, f_2$  independent; then  $\text{cov}(f, f') = k_1 + k_2$ . Useful for combining kernels with different characteristic length-scales.

**Product.**  $k = k_1 k_2$  is a valid kernel. *Proof:* take  $f(x) = f_1(x) f_2(x)$  with  $f_1, f_2$  independent; then  $\text{cov}(f, f') = k_1 k_2$ . A simple extension shows  $k^p$  is valid for any  $p \in \mathbb{N}$ .

**Scaling.** For deterministic  $a(x)$  and  $g(x) = a(x)f(x)$ :

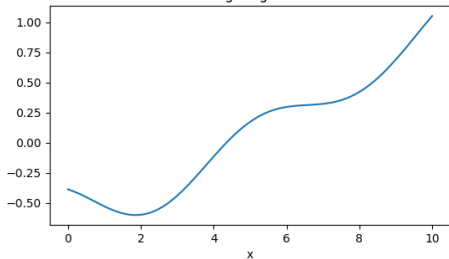
$$\text{cov}(g(x), g(x')) = a(x) k(x, x') a(x').$$

Choosing  $a(x) = k(x, x)^{-1/2}$  **normalizes** the kernel:

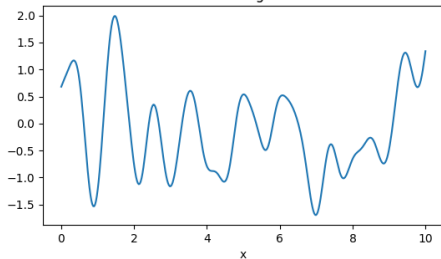
$$\tilde{k}(x, x') = \frac{k(x, x')}{\sqrt{k(x, x) k(x', x')}}, \quad (4.35)$$

so that  $\tilde{k}(x, x) = 1$  for all  $x$ .

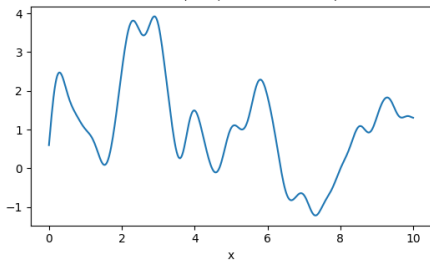
k1: long length-scale



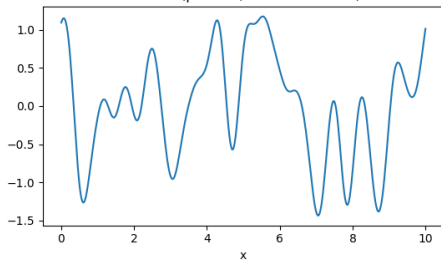
k2: short length-scale



$k_1 + k_2$  (sum, R&W Sec. 4.2.4)



$k_1 * k_2$  (product, R&W Sec. 4.2.4)



# Making New Kernels from Old: Convolution & Products of Spaces

**Convolution (blurring).** For a fixed kernel  $h(x, z)$  and  $g(x) = \int h(x, z) f(z) dz$ :

$$\text{cov}(g(x), g(x')) = \iint h(x, z) k(z, z') h(x', z') dz dz'.$$

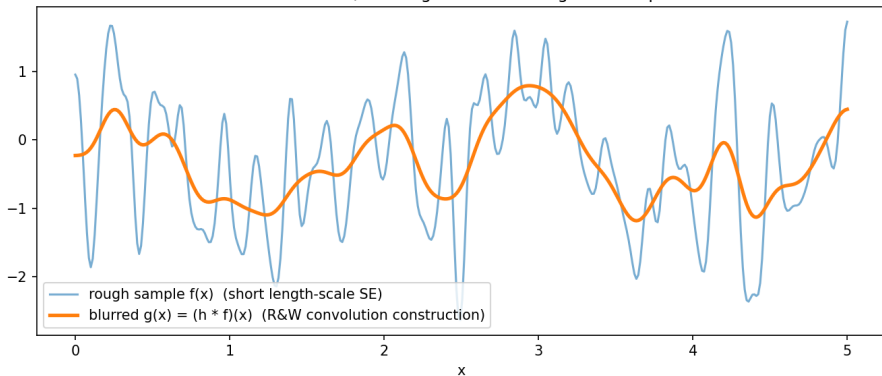
**Combining different input spaces.** If  $k_1(x_1, x'_1)$  and  $k_2(x_2, x'_2)$  are kernels on spaces  $X_1, X_2$ , then on the product space  $X_1 \times X_2$ :

$$\text{direct sum: } k = k_1(x_1, x'_1) + k_2(x_2, x'_2)$$

$$\text{tensor product: } k = k_1(x_1, x'_1) k_2(x_2, x'_2)$$

are both valid immediate from the sum and product rules above.

Convolution / blurring smooths a rough GP sample



# Research Gaps: Non-Stationary Kernels in High Dimensions

## ① Automated kernel search lacks non-stationary constructions

- ▶ Restricted to *sums/products* of stationary base kernels - nothing beyond it.
- ▶ “*The role of non-stationary kernels in high-dimensional BO warrants further investigation.*”

Yun, Shin, Song, Lee & Park (2026). *Automated Kernel Discovery Towards Understanding High-Dimensional Bayesian Optimization*. arXiv:2605.20249.

## ② No general- $D$ Gibbs/Paciorek–Schervish implementation

- ▶ convoSPAT/BayesNSGP: 2D only. hetGP: scales, but only variance is non-stationary.

Booth, Cooper & Gramacy (2023, rev. 2024). *Nonstationary Gaussian Process Surrogates*. arXiv:2305.19242.

## ③ Kernel-hyperparameter uncertainty ignored in experimental design

- ▶ Point-estimated hyperparameters discard posterior uncertainty from the acquisition function.

Jones & Dowling (2026). *BITS for GAPS: Bayesian Information-Theoretic Sampling for Hierarchical GAussian Process Surrogates*. arXiv:2511.16815.